Walnuts improve semen quality in men consuming a Western-style diet: randomized control dietary intervention trial.

We tested the hypothesis that 75 g of whole-shelled walnuts/day added to the Western-style diet of healthy young men would beneficially affect semen quality. A randomized, parallel two-group dietary intervention trial with single-blind masking of outcome assessors was conducted with 117 healthy men, age 21-35 yr old, who routinely consumed a Western-style diet. The primary outcome was improvement in conventional semen parameters and sperm aneuploidy from baseline to 12 wk. Secondary endpoints included blood serum and sperm fatty acid (FA) profiles, sex hormones, and serum folate. The group consuming walnuts ($n = 59$) experienced improvement in sperm vitality, motility, and morphology, but no change was seen in the group continuing their usual diet but avoiding tree nuts ($n = 58$). Comparing differences between the groups from baseline, significance was found for vitality ($P = 0.003$), motility ($P = 0.009$), and morphology (normal forms; $P = 0.04$). Serum FA profiles improved in the walnut group with increases in omega-6 ($P = 0.0004$) and omega-3 ($P = 0.0007$) but not in the control group. The plant source of omega-3, alpha-linolenic acid (ALA) increased ($P = 0.0001$). Sperm aneuploidy was inversely correlated with sperm ALA, particularly sex chromosome nullisomy (Spearman correlation, -0.41, $P = 0.002$). Findings demonstrated that walnuts added to a Western-style diet improved sperm vitality, motility, and morphology.